



Normal Microbiota of the GI Tract

Microbiology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. Which gut segment has the densest bacterial population, and roughly how many per mL?
2. Why is the stomach almost empty of bacteria, and which organism persists there?
3. What happens to *C. difficile* when broad-spectrum antibiotics wipe out the normal flora?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Definitions and how it starts

Term	Meaning
Microbiota	Microorganisms (bacteria, fungi, viruses) on skin and mucous membranes of healthy people; growth depends on temperature, moisture, nutrients
Microbiome	The collective genomes of these microbial symbionts
Resident	Relatively fixed types in each site; re-establish promptly if disturbed
Transient	From the environment; non-pathogenic or potentially pathogenic; not permanent ; little significance while residents are intact — may colonise and cause disease if the resident flora is disturbed
Early life	Environment (maternal vaginal, fecal, skin flora) shapes the early profile; breast-fed → Bifidobacteria dominate ; bottle-fed → more mixed, fewer lactobacilli

2 Along the tract — bacterial density climbs



The colon holds the highest count of any microbial habitat. In diarrhea counts fall; in intestinal stasis they rise.

Site	Key points
Esophagus	Microbes arrive with saliva and food
Stomach	Low density because of gastric enzymes and acid pH ; of hundreds of phylotypes only <i>Helicobacter pylori</i> persists ; acid protects against enteric pathogens (e.g. <i>V. cholerae</i>); antacids, H₂ blockers and PPIs → marked rise in flora , incl. faecal organisms
Small intestine	Flora rises as pH becomes alkaline; upper-intestine mucosa: Bacteroidetes and Clostridiales
Lumen / cecum	Enterobacterales and enterococci
Colon	Sigmoid / rectum: bacteria ≈ 60% of fecal mass ; anaerobes outnumber facultative organisms 1000-fold ; 96-99% anaerobes ; six phyla: Bacteroidetes, Firmicutes, Actinobacteria, Verrucomicrobiota, Fusobacteria, Proteobacteria
Others in the colon	Archaea (<i>Methanobrevibacter smithii</i> , methane producer); protozoa and fungi; phages are the commonest viruses
Clinical hook	Minor trauma (sigmoidoscopy, barium enema) → transient bacteraemia in ~10% of procedures

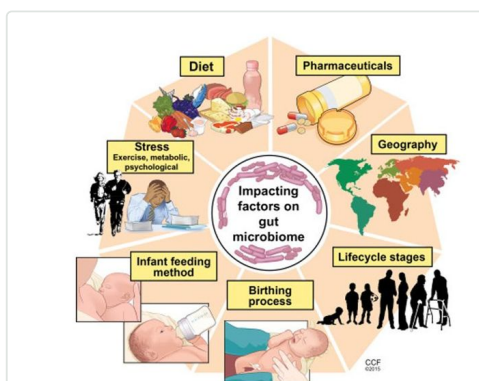
3 Bacteroidetes at a glance (lecture table)

Species	Role
<i>B. fragilis</i>	Pathogenic — <i>B. fragilis</i> toxin (BFT); multidrug-resistant infections
<i>B. thetaiotaomicron</i>	Anti-inflammatory; B ₁₂ transporter; suppresses <i>C. difficile</i> toxin ; promotes gut-barrier regeneration
<i>B. vulgatus</i>	Inflammatory (↑ osteoclast activity)
<i>B. uniformis</i>	Antilipogenic (degrades plant polysaccharides)
<i>B. ovatus</i>	Immune support; GABA and SCFA production

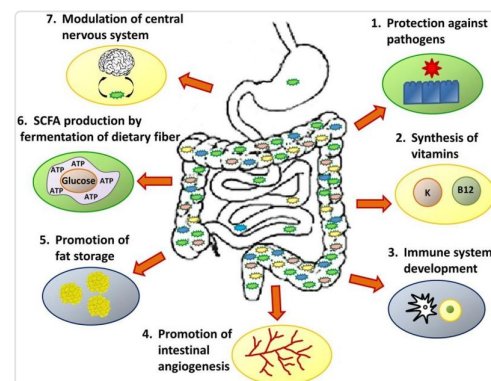
<i>Prevotella copri</i> · <i>Parabacteroides distasonis</i> · <i>Alistipes finegoldii</i>	Immune support (SCFA except propionate) · anti-inflammatory · protective against colitis
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4 What shapes the flora

Factor	Effect
Diet	Vegetarian → ↑ saccharolytic organisms (e.g. <i>E. coli</i>); meat → ↑ proteolytic (<i>Clostridium</i>). Animal-based diet → ↑ bile-tolerant <i>Alistipes</i> , <i>Bilophila</i> , <i>Bacteroides</i> ; ↓ Firmicutes that ferment plant polysaccharides (<i>Roseburia</i> , <i>Eubacterium rectale</i> , <i>Ruminococcus bromii</i>)
HCl and bile	Antimicrobial → fewer microbes in stomach and duodenum
Physiology	Diarrhea washes microbes out
Birth and infancy	Mode of delivery (vaginal vs caesarean); breast milk vs formula
Antibiotics	Oral antibiotics ↓ numbers; prolonged courses markedly ↓ flora



Factors impacting gut microbiome (lecture figure).



Functions of intestinal microbiota.

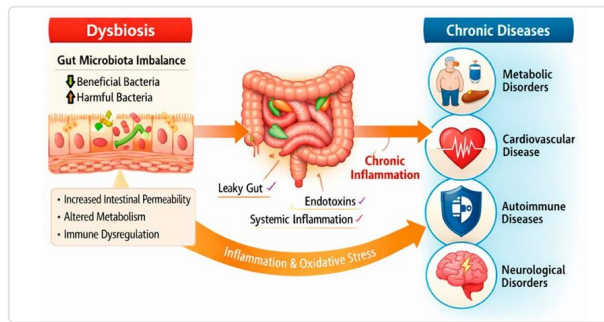
5 What the flora does

Function	Detail
Colonisation resistance	Displace and inhibit pathogens: compete for nutrients and receptors; make bacteriocins and lactic acid
Immune development	Needed for the mucosal immune system ; induce IgA ; "teach" the immune system to react to microbial antigens
Metabolism	Small-bowel flora can supply amino acids; make vitamin K, biotin, folate ; enhance ion absorption; metabolise dietary carcinogens; ferment non-digestible residue (SCFA)

6 Symbiosis, dysbiosis and disease

	Meaning
Symbiosis	The type of interaction between organisms (beneficial balance of commensals and pathogens)
Homeostasis	The stable condition that can result from symbiosis
Dysbiosis	Imbalance in number or type of colonising microbes

- **C. difficile:** lives in tiny numbers, held in check by microbial antagonism; **broad-spectrum / long-term antibiotics** let it flourish → severe, long-lasting intestinal disease
- **Bowel perforation:** colonic anaerobes (*B. fragilis*, clostridia, peptostreptococci) drive **abscess formation**
- **IBD** (and possibly colon cancer): in genetically susceptible people, loss of immune tolerance to bacterial antigens → exuberant inflammation
- Stress, excess alcohol, smoking, antibiotics, unhealthy diet → ↑ pathogens, ↓ commensals; dysbiosis → leaky gut → endotoxins → chronic disease (metabolic, cardiovascular, autoimmune, neurological)



Gut dysbiosis and chronic disease (lecture figure).

Fecal microbiota transplantation (FMT) Detail	
What	Transfer of fecal bacteria from a healthy donor into a recipient (stool transplant)
Use / logic	Successful for <i>C. difficile</i> ; bacterial interference — harmless bacteria displace pathogens; restores missing Bacteroidetes and Firmicutes

P Must-remember pearls

- Density: **stomach** 10^2 - 10^3 → **colon** 10^{11} - 10^{12} — the highest of any habitat; colon is **96-99% anaerobes**.
- Stomach acid keeps counts low; only ***H. pylori*** persists; **PPIs / H_2 blockers / antacids** raise gastric flora.
- Functions: **displace pathogens**, develop mucosal immunity (**IgA**), make **vitamin K, biotin, folate**.
- **Symbiosis** = interaction; **homeostasis** = the stable result; **dysbiosis** = imbalance.
- Antibiotics → ***C. difficile*** overgrowth; **FMT** restores Bacteroidetes and Firmicutes.

Q Test yourself

1. A man on long-term omeprazole has bacterial overgrowth in the stomach, including organisms normally found in feces. Why?

- A. Loss of gastric acid removes a key barrier
 B. Increased salivary IgA
 C. Excess bile in the stomach
 D. Increased peristalsis

2. A 70-year-old with sigmoid diverticulosis has a ruptured diverticulum and a pericolic abscess. Most likely bacteria?

- A. *B. fragilis* alone
 B. *E. coli* alone
 C. Mixed normal GI flora
 D. *Enterococcus* alone

3. The stable condition that results from a balanced host-microbe interaction is called:

- A. Homeostasis
 B. Dysbiosis
 C. Colonisation resistance
 D. Microbial antagonism

4. A patient given prolonged antibiotics develops severe diarrhea. Which organism can proliferate and produce toxin?

- A. *Enterococcus*
 B. *S. epidermidis*
 C. *Clostridioides difficile*
 D. *B. fragilis*

5. Which organism dominates the gut of a breast-fed infant?

- A. *Bifidobacteria*
 B. *Clostridium*
 C. *Bacteroides*
 D. *Enterobacterales*

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — A** — Acid and gastric enzymes keep stomach counts low; PPIs remove that barrier.
2 — C — Colonic anaerobes and facultative organisms together form the abscess after perforation.
3 — A — Symbiosis is the interaction; homeostasis is the stable result.
4 — C — Loss of diverse normal flora removes antagonism, letting *C. difficile* flourish.
5 — A — Breast-fed infants are dominated by Bifidobacteria; bottle-fed infants have a more mixed flora.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam